

Health Intelligence Report

Patient: Rachel Bertler
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NOTICE: This report identifies patterns in health data to support informed conversations with a healthcare provider. It is not a medical diagnosis. Always consult a qualified clinician.

SUMMARY

This health profile shows a cluster of mental health diagnoses (burnout, depression, sleep difficulties) emerging in April 2026, alongside persistently low vitamin D levels (14.8 ng/mL in July 2025, well below the 30 ng/mL threshold) that have remained deficient for over 5 years. Carbon dioxide levels have also trended downward into the flagged range. The long-standing endometriosis diagnosis may interact with current symptoms, particularly fatigue and mood.

PATTERNS IDENTIFIED

Chronic Vitamin D Deficiency Coinciding with Mental Health Symptoms

Confidence: Strong pattern

RELATED CONDITIONS

Major Depressive Disorder · Burnout: occupational · Sleep Difficulties

SUPPORTING EVIDENCE

- Vitamin D at 14.8 ng/mL, significantly below reference range (30–100 ng/mL) — Lab Results, July 2025
- Vitamin D deficiency persisted from 10 ng/mL in March 2020 with minimal improvement over 5+ years — Lab Results, March 2020 to July 2025
- Major Depressive Disorder, Burnout, and Sleep Difficulties all diagnosed — Health History, April 2026

Declining Carbon Dioxide Levels May Indicate Respiratory or Metabolic Pattern

Confidence: Possible pattern

RELATED CONDITIONS

Sleep Difficulties · Burnout: occupational · Major Depressive Disorder

SUPPORTING EVIDENCE

- Carbon Dioxide dropped from 20 mmol/L to 19 mmol/L (below reference range of 20–29) — Lab Results, March 2020 to July 2025
- Sleep Difficulties diagnosed — Health History, April 2026
- Burnout and depression present, which can affect breathing patterns — Health History, April 2026

Endometriosis as Potential Contributor to Chronic Stress and Mood Impact

Confidence: Possible pattern

RELATED CONDITIONS

Endometriosis: minimal · Burnout: occupational · Major Depressive Disorder

SUPPORTING EVIDENCE

- Minimal endometriosis diagnosed and ongoing since August 2007 — Health History, August 2007
- Burnout and Major Depressive Disorder diagnosed 19 years later — Health History, April 2026

INFORMATION GAPS

- Thyroid panel (TSH, Free T4) to rule out hypothyroidism as a contributor to fatigue, depression, and sleep difficulties
- Complete metabolic panel including electrolytes to contextualize the low CO2 trend
- Iron studies and ferritin, particularly given endometriosis history and potential for chronic blood loss contributing to fatigue
- Most recent mental health treatment details (medications, therapy) to assess whether vitamin D or other physiological factors have been addressed
- Sleep study or sleep quality assessment to investigate whether sleep difficulties involve apnea or other respiratory patterns that might explain low CO2

TALKING POINTS FOR YOUR PROVIDER

- Should we explore aggressive vitamin D supplementation (likely higher than standard doses given the persistent deficiency) and reassess mood and sleep symptoms after levels normalize?
- Could the low CO₂ be related to hyperventilation from anxiety/stress, or should we rule out sleep-disordered breathing given the sleep difficulties diagnosis?
- Has the endometriosis been re-evaluated recently, and could chronic pain or inflammation be contributing to burnout and depression rather than solely occupational factors?

